

Resonant scattering effects of the members in Maria family

Introduction

illustrated by Robert Cha

Inner Solar System



Where do NEAs come from?

Robert

We are from the Main Belt.



NEA



How did you get here?

We get here with spinning!



NEA



How did you do that?

Let me tell you the story!



NEA

When I was in my hometown, I was orbiting with my family together

Why are they so noisy?



Mars

Guys, let's orbit!



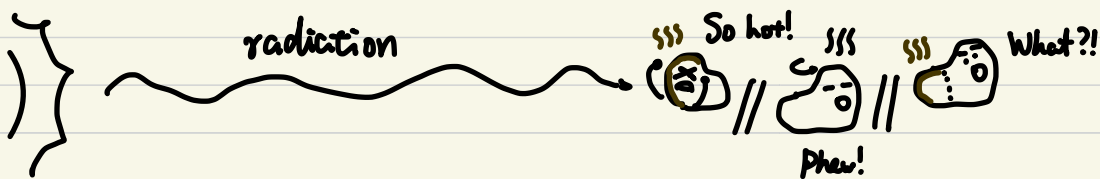
MBA

I don't know.



Jupiter

However, once a magic called "radiation" from the center lightened me! At that time, my head got fever, so I turned around to get cooler. Unfortunately, my tail began with heating soon.



Thus, keeping trunning around, I lost my family unconsciously. When I looked around, I was approaching Mars. And then, I saw the Earth nearby eventually. That's why I'm here today.

Hello, are you okay?



Mars



MBA
MCA



Earth

MCA
NEA

Earth, someone's coming!



Mars

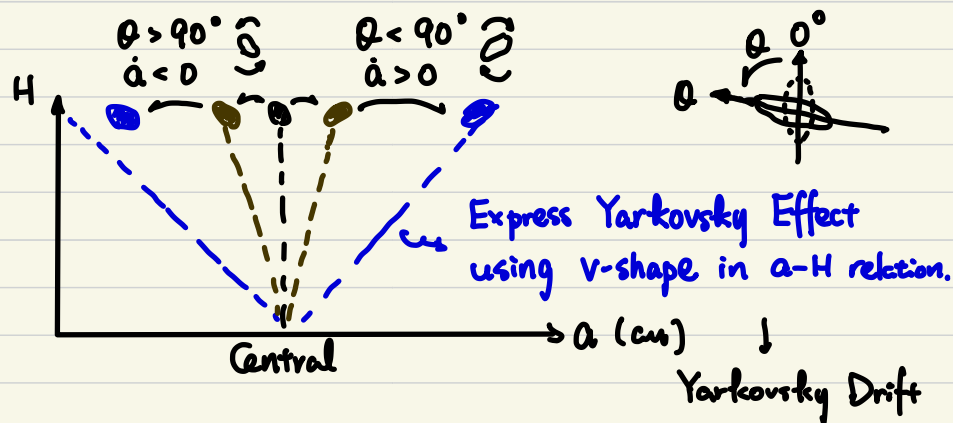
I see. That's an upset story, but I think there's an effect suscribing you said above.
Oh I remember that! That calls "Yarkovsky Effect".

Yarkovsky Effect

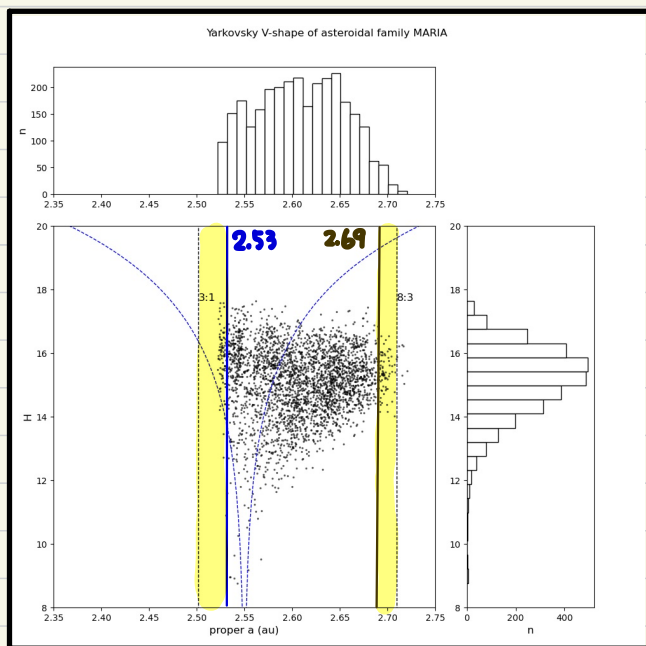
I heard David Jewitt, 2025 mentioning that

as a non-gravitational force, Yarkovsky Effect is occured with solar radiation.
It will heat the part of facing Sun. Different temperature at the same body will spin continuously.

solar radiation



I think that's why you are here. Luckily, I'm investigating an elder asteroid family, Maria family, an S-type family with high inclination ($12^\circ \sim 17^\circ$).



The primary reason causing these asteroids break away their family is due to nearby resonant area. So I select some members around 3:1 and 8:3 resonance.

Calculation of Yarkovsky drift

 Bryce T. Bolin⁺, 2018^[2]

$$\frac{da}{dt}(D, \alpha, a, e, A) = \left[\frac{da}{dt} \right]_0 \frac{\sqrt{a_0}(1-e_0^2)}{\sqrt{a}(1-e^2)} \left[\frac{D_0}{D} \right]^\alpha \cdot \left[\frac{\rho_0}{\rho} \right] \left[\frac{1-A}{1-A_0} \right] \left(\frac{au}{Myr} \right) \cdot \frac{\cos \theta}{\cos \theta_0}$$

↳ At zeropoint in inner main belt

	Maria's	Zeropoint	Unit	Remark
Zeropoint Drift $(da/dt)_0$		$4.7 \cdot 10^{-5}$	au/Myr	inner main belt
Semi-major axis (a)	2.55	2.37	au	Central for Maria family
Eccentricity (e)	~ 0.1	0.2		
Diameter sized (D)	1	5	km	Assumption for Maria family
Density (ρ)	2600	2500	kg/m ³	General S-type asteroids
Albedo (A)	0.2609	0.1		Maria from Aljbaae ⁺ , 2017 ^[3]
Obliquity (θ)	$\begin{cases} 180 & (3:1) \\ 0 & (8:3) \end{cases}$	0	deg	



Put these values in the table to the Equation on top.

$$\left[\frac{da}{dt} \right]_{1\text{-km Maria}} \approx 4.7 \cdot 10^{-5} \cdot \frac{\sqrt{2.37} \cdot [1-(0.2)^2]}{\sqrt{2.55} \cdot [1-(0.1)^2]} \cdot \left(\frac{5}{1} \right)^{0.8} \cdot \left(\frac{2500}{2600} \right) \cdot \left(\frac{1-0.1}{1-0.2609} \right) \cdot \frac{\cos \theta}{1} \frac{au}{Myr}$$

$$\sim \begin{cases} -2 \cdot 10^{-4} \text{ au/Myr} & \dots \theta = 180^\circ \text{ for } 3:1 \text{ MMR} \\ 2 \cdot 10^{-4} \text{ au/Myr} & \dots \theta = 0^\circ \text{ for } 8:3 \text{ MMR} \end{cases}$$

Determine v-shape^[2]

The standard case V-shape in a vs. $1/D$

$$\Delta a \equiv a - a_c = da/dt(D) \cdot \Delta t \rightarrow \text{age}$$

Consider order α . So we define the const. C and C_α ,

$$C = \Delta t \cdot \left[\sqrt{A} \cdot \left(\frac{da}{dt} \right)_0 \right]$$

$$= C_\alpha \cdot \left[\frac{\sqrt{A}}{1329} \right]^{1-\alpha}$$

Hence,

$$\frac{1}{D} = \frac{\sqrt{A}}{1329} \left[\frac{|\Delta a|}{C_\alpha} \right]^{1/\alpha}$$

$$= \left[\frac{|\Delta a| \sqrt{A}}{1329 C} \right]^{1/\alpha}$$

In absolute magnitude term (H) ,

$$\frac{1}{D} = \left(\frac{1329}{\sqrt{A}} \right)^{-1} \cdot 10^{H/5}, \quad H = 5 \cdot \log_{10} \left[\left(\frac{|\Delta a|}{C} \right)^{1/\alpha} \cdot \left(\frac{\sqrt{A}}{1329} \right)^{1/\alpha - 1} \right]$$

To be continued ~